

Erythema Toxicum Neonatorum (ETN)

Erythema toxicum neonatorum (ETN) is an idiopathic, common condition seen in up to 75 percent of term newborns. It is rarely seen in premature infants. Blotchy erythematous macules 1 to 3 cm in diameter with a 1- to 4-mm central vesicle or pustule are characteristic of ETN. These follicular-based lesions typically appear at 24 to 48 hours of age, although delayed onset up to 10 days has been reported. The lesions can occur anywhere on the body but tend to spare the palms and soles.

A smear of the central vesicle or pustule contents shows numerous **eosinophils** on Wright-stained preparations. Peripheral blood eosinophilia (up to 20%) may be present, especially in infants with widespread lesions.

Transient neonatal pustular melanosis (TNPM) may resemble ETN, but it is distinguished by the presence of **neutrophils** rather than eosinophils in the vesicles, and by the development of residual pigmentation at lesion sites—features not seen in ETN.

Other conditions that may mimic ETN include: - Bacterial infections - *Pityrosporum* folliculitis - Congenital candidiasis

Differentiation can be achieved through bacterial and fungal cultures and Gram staining of lesional material.

ETN is a **benign condition** that resolves spontaneously by 2 to 3 weeks of age without scarring or sequelae.

Transient Neonatal Pustular Melanosis (TNPM)

TNPM is an idiopathic pustular eruption of newborns that heals with tiny brown pigmented macules. It is less common than ETN and occurs more frequently in newborns with **darker skin pigmentation**.

Lesions are present **at birth or appear shortly thereafter**, presenting as superficial vesicles and pustules. Ruptured lesions may show collarettes of scale.

Pigmented macules are often present at birth or develop rapidly at the sites of resolving pustules within hours to the first day of life.

The distribution includes the forehead and mandibular area, and unlike ETN, **palms and soles may be involved.**

A smear of vesicle or pustule contents reveals a **predominance of neutrophils**, with occasional eosinophils on Wright-stained preparations.

TNPM is a **harmless condition** requiring no treatment. Pustules usually resolve by 5 days of age, leaving pigmented macules that fade over 3 weeks to 3 months.

Miliaria Rubra

Miliaria rubra is often confused with ETN and TNPM. However, the surrounding erythema is smaller (1–2 mm) compared to the 20–30 mm macules seen in ETN.

The central pustule of TNPM may resemble: - Congenital candidiasis - *Pityrosporum* folliculitis - Bacterial folliculitis

Herpes simplex virus infection should be considered if vesicular lesions are present.

A **Gram-stained slide** of pustule contents in ETN or TNPM will **not show organisms.** A Wright-stained slide typically shows a predominance of **neutrophils.**

Mottling

Mottling refers to a blotchy or lace-like pattern of dusky erythema on the extremities and trunk, commonly triggered by cold exposure. It is a **physiologic phenomenon** due to immature autonomic regulation of the cutaneous vascular plexus and is observed in nearly all newborns at some point.

It **resolves with warming** and typically disappears spontaneously by 6 months of age. This helps differentiate it from pathologic conditions such as **cutis marmorata telangiectatica congenita** and **livedo reticularis**.

Harlequin Color Change

Harlequin color change is a rare, transient vascular phenomenon seen primarily in **low-birth-weight infants**. It is characterized by a striking, sharp demarcation of erythema on one side of the body, with pallor on the opposite side, often occurring during sleep or in response to external stimuli. The condition is benign and resolves spontaneously within minutes to hours.